Kriti Goomer

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OBJECTIVE

Second-year B.S. Computer Science student (4.0 GPA) at UT Dallas with full stack production experience (React/Node/Firebase) serving 900+ users, seeking software engineering or ML/cloud internship. Strong interest in AI technologies, distributed systems, and building production grade applications that integrate machine learning with scalable cloud infrastructure.

EDUCATION

The University of Texas at Dallas May 2027
Bachelor of Science, Computer Science GPA: 4.0

- **Honors:** Dean's List (Fall 2024, Spring 2025, Fall 2025), Academic Excellence Scholarship
- **Relevant Coursework:** Probability & Statistics for CS, Linear Algebra, Advanced Algorithm Design & Analysis, Systems Programming in UNIX (C), Computer Architecture (MIPS Assembly), Software Engineering, Data Structures & Algorithms (Java), Discrete Mathematics

EXPERIENCE

Software Developer | Artificial Intelligence Society (AIS), UT Dallas September 2025-Present

- Led design and implementation of the *HackAI* (927 applicants, 364 participants) and *AI Mentorship* pages (500+ users)
- Built dynamic frontend using **Next.js**, **React**, **TypeScript** with **Firebase Firestore** for real-time data and role-based access
- Engineered **Node.js** scripts to migrate and normalize 927 application records from **Coda** to **Firebase** with data integrity
- Implemented authentication, access control, and automated secure access code generation
- Collaborated on a 6-developer team using **Git** workflows to scale systems serving 400+ active members

Front-End Developer | *Comet Connect*, Society of Women Engineers (SWE), UT Dallas September 2024-May 2025

- Designed and implemented mobile UI/UX for a campus networking application using **React Native**
- Built **Firebase authentication** workflows including secure user login and protected data routing
- Developed real-time messaging and user filtering features to improve user engagement and retention
- Presented system architecture and implementation strategy to industry engineers

PROJECTS

BabySteps - Real-Time Contraction Monitoring System | React Native, Node.js, Arduino, WebSockets March 2026

- Built an end-to-end real-time monitoring system integrating **Arduino** sensor data with a **React Native** mobile app via a **Node.js** backend
- Developed a **WebSocket** pipeline to stream contraction events (start/end, duration) with low-latency updates to the frontend
- Implemented a serial-to-WebSocket bridge for communication between Arduino (**C++**) and client applications
- Designed session tracking and contraction classification logic with real-time analytics (intervals, durations, labor likelihood)
- Integrated device motion detection (**Expo**) and calibration flow to improve signal reliability and reduce noise

Concurrent Client-Server Messaging System | C, Linux, Threads, Processes, Sockets Fall 2025

- Implemented TCP multi-client messaging server in C using POSIX sockets and concurrent process management
- Designed client connection handling with multiplexed I/O
- Ensured thread-safe message broadcasting across active clients

Browser Navigation Engine | Java Fall 2025

- Designed dual-stack-based browser history model support forward/back navigation of core operations in **Java**
- Incorporated state saving across sessions/runs and transition logic to invalidate forward history on new navigation

Anthropic Katha June 2023-August 2023

- Built educational platform explaining particle physics concepts
- Reached 300+ global users; simplified complex STEM topics via structured content

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript
Frontend: React, React Native, HTML, CSS/Tailwind
Backend & Cloud: Node.js, Express, WebSockets, REST APIs, Firebase (Firestore, Auth), TCP/IP, Serial Communication, GCP
Tools & Systems: Git/GitHub, Linux, Socket Programming, Multithreading/Concurrency, Embedded Systems (Arduino), Real-Time Data Streaming, Figma
ML/Data: Pandas, NumPy, Scikit-learn